YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR

(An Autonomous Institution Affiliated to RashtrasantTukadoji Maharaj Nagpur University)

DEPARTMENT OF INFORMATION TECHNOLOGY

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LMS: Learning Management System using LlaMa and SearXNG

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Abstract: Indian students often face challenges with online content overload and the lack of structured learning paths. This project introduces an innovative Learning Management System (LMS) leveraging a locally run Llama language model via Ollama and the privacy-respecting metasearch engine SearXNG. The system features a decision agent for intelligent query routing, automated topic decomposition to create personalized learning roadmaps, and curated video recommendations with an adaptive user feedback loop. By providing contextually relevant information and structured guidance while ensuring data privacy, this LMS aims to streamline the learning process and enhance educational outcomes for students navigating vast digital resources.

Keywords : Learning Management System (LMS), LLaMa, SearXNG, Ollama, Artificial Intelligence (AI), Personalized Learning, Adaptive Learning, Educational Technology (EdTech), Topic Decomposition, Video Recommendation, Decision Agent, User Feedback Loop, Privacy-Preserving Search, React, Express.js, Node.js, Large Language Models (LLMs)

Introduction: The digital era presents an unprecedented volume of educational content, often leading to information overload and a lack of clear direction for students, a challenge acutely felt in regions like India. To address this, our project introduces an innovative Learning Management System (LMS) designed to streamline and personalize the learning journey. This system integrates the power of advanced Large Language Models, specifically Llama (run locally via Ollama to ensure user privacy), with SearXNG, a privacy-respecting metasearch engine for accessing relevant, real-time information. Built with a React frontend and an Express.js backend, the LMS employs a sophisticated decision agent to intelligently route user queries. Key features include AI-driven decomposition of complex topics into manageable sub-topics, followed by the curation of educational video recommendations.

Stimulated Diagram:

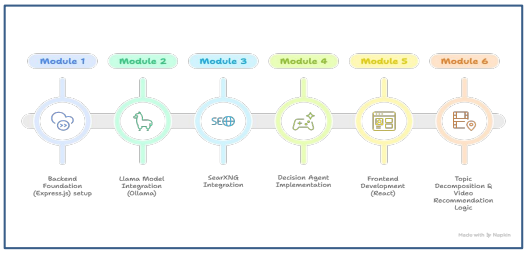


Fig. 1: Project Modules

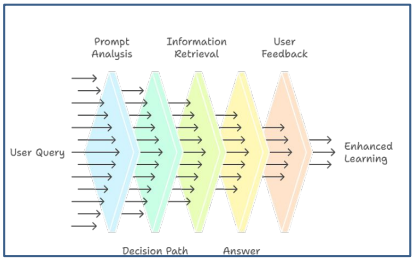


Fig. 2: Workflow of Project

Result:

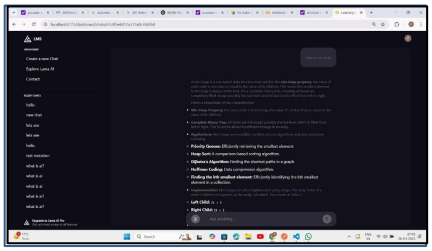


Fig.3 Content Generation

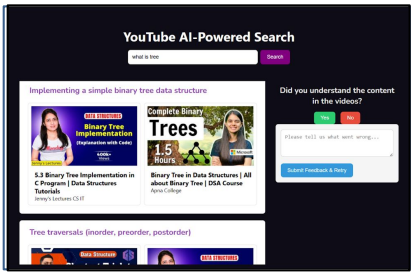


Fig.4: Youtube Search Feature

Conclusion :

This Learning Management System, integrating locally-run Llama models via Ollama and the privacy-focused SearXNG, successfully tackles content overload for Indian students. By offering AI-driven topic decomposition, personalized learning roadmaps, curated video recommendations, and an adaptive feedback loop, it provides a structured and secure educational experience.

Future Scope:

Future scope includes deeper adaptive learning paths, LLM-generated interactive content like quizzes and summaries, expanded multilingual support, and educator dashboards with detailed learner analytics for continuous system and learning enhancement.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	3	3			3	3	3		3	3
CO2					3	3	3	3	3	3	3	3	3
CO3		3	3	3								3	3
CO4					3				3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3

Signature Of Guide: _____